PROJECT REPORT

DOOR STATUS INDICATOR

Schematic Design and PCB Implementation

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1. Schematic Diagram

The schematic represents the electrical connections for the Door Status Indicator module. The design includes a two-pin connector, two resistors, red and green LEDs, an SPST switch, and a buzzer. This schematic is the electrical reference for the PCB layout.

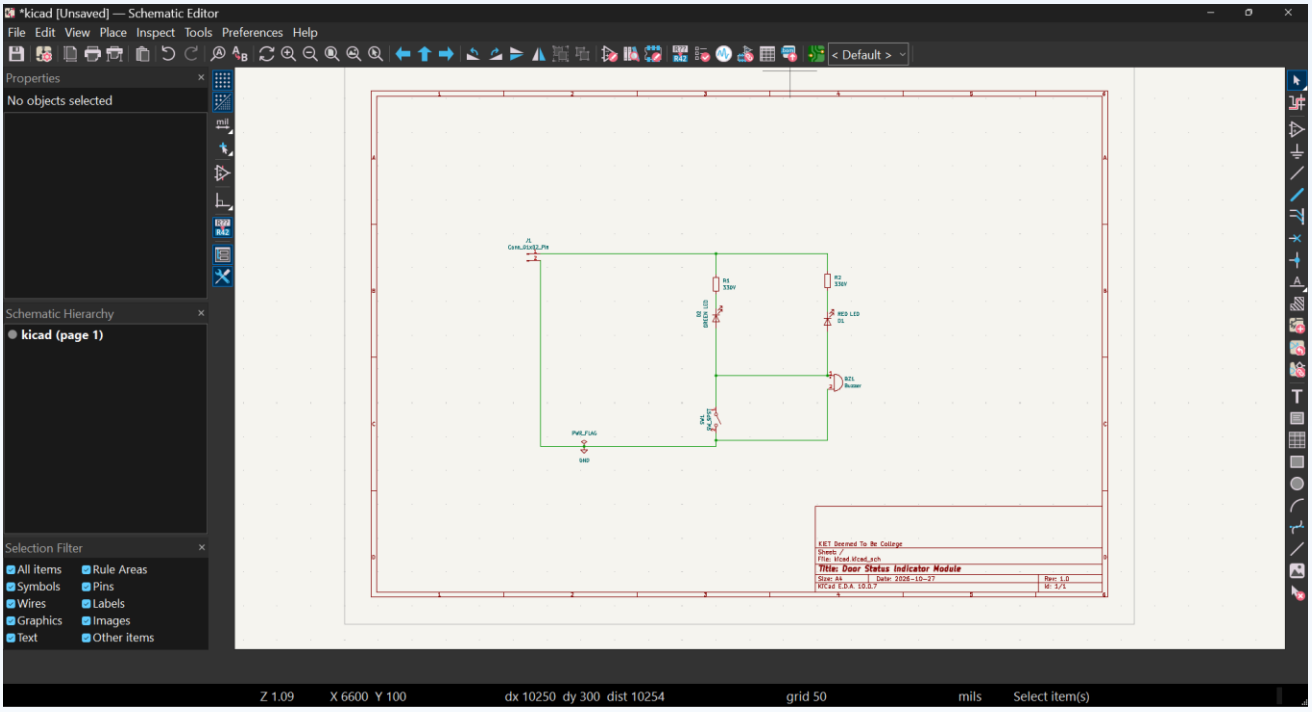


Figure 1. Circuit schematic captured in KiCad Schematic Editor.

2. Footprint Assignment

The footprint-assignment step maps each schematic symbol to the physical package used on the PCB. The screenshot below shows the assignments selected in KiCad.

Ref.	Component	Assigned footprint
BZ1	Buzzer	Buzzer_Beeper:Buzzer_12x9.5RM7.6
D1	Red LED	LED_THT:LED_D5.0mm
D2	Green LED	LED_THT:LED_D5.0mm
J1	2-pin connector	Connector_PinHeader_2.54mm:PinHeader_1x02_P2.54mm_Vertical
R1, R2	Resistors	Resistor_THT:R_Axial_DIN0207_L6.3mm_D2.5mm_P7.62mm_Horizontal
SW1	SPST switch	Button_Switch_THT:SW_SPST_Omron_B3F-40xx

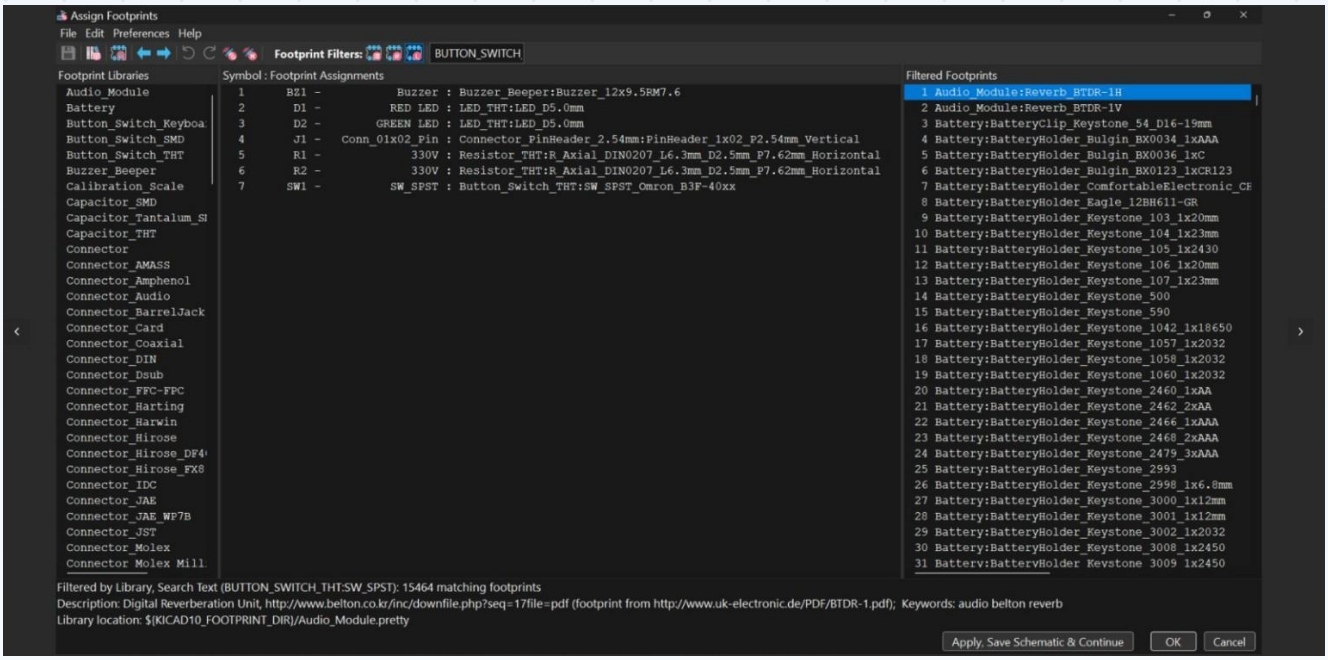


Figure 2. KiCad Assign Footprints window.

Check before fabrication: the resistor value appears as “330V” in the screenshot. Confirm the intended resistance and verify each footprint against the actual component datasheet.

3. 2D PCB View — Routed Board

The 2D PCB view shows the board outline, component placement, pads, and copper tracks. The supplied KiCad screenshot shows 18 pads, 27 track segments, 5 nets, and 0 unrouted connections in the displayed status bar.

6. 2D PCB View - Routed Board

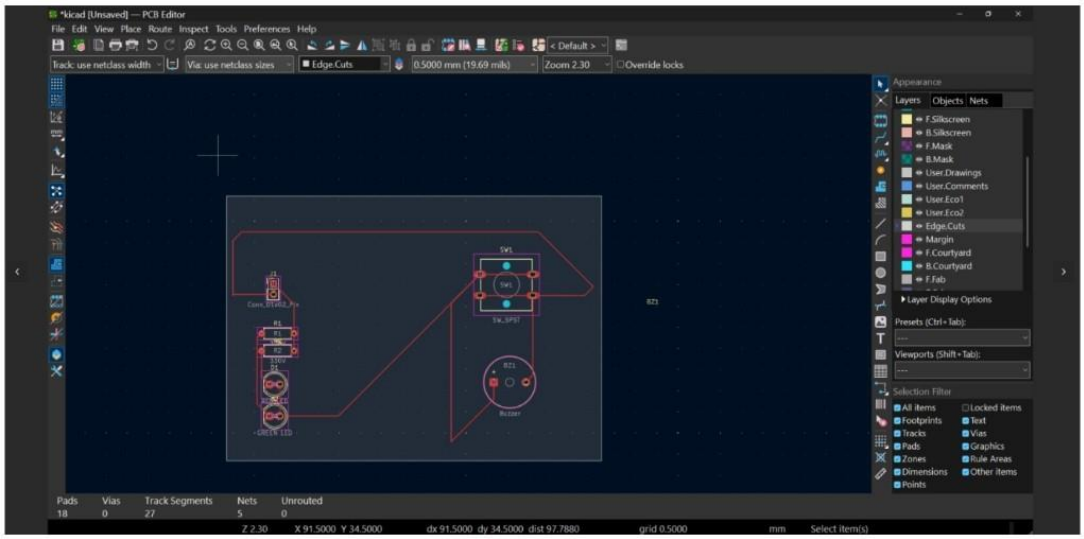


Figure 3. Routed PCB layout in KiCad PCB Editor.

4. 3D PCB View

The 3D preview provides a visual check of component placement, board shape, and orientation. The board shown includes the connector, two resistors, two LEDs, a switch, and a buzzer.

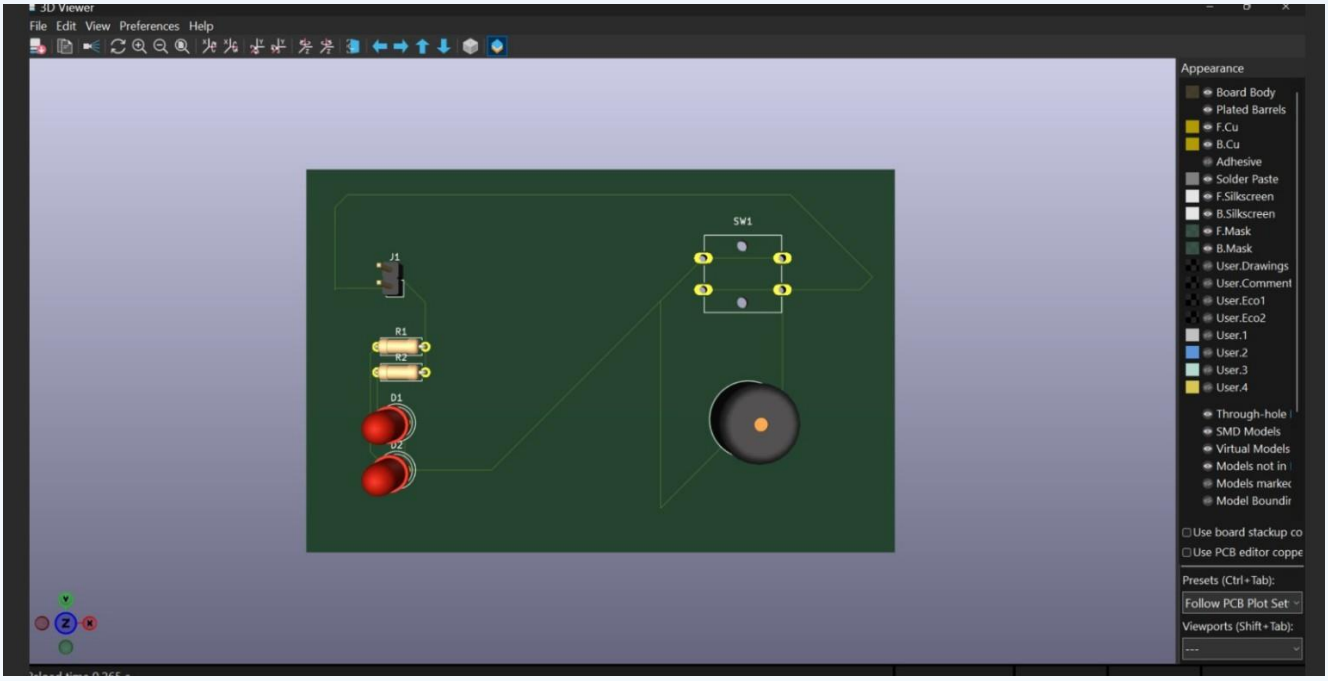


Figure 4. Three-dimensional PCB preview in KiCad 3D Viewer.

5. Working Principle

The Door Status Indicator is intended to give a visible and audible indication of a door's state. The switch acts as the state input. Depending on the switch position and the final circuit wiring, the red and green LEDs provide visual indication while the buzzer provides an audible alert. The resistors limit current through the LEDs.

Before powering the circuit, check the supply voltage, LED polarity, buzzer rating, switch connections, and resistor values. Confirm all electrical connections and run the PCB design-rule check before fabrication.

6. Conclusion

This project demonstrates the PCB design workflow in KiCad, from schematic capture and footprint assignment to 2D routing and 3D visualization. The included screenshots document the key stages of the design. Final assembly should proceed only after checking component compatibility, electrical values, and board clearances.

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